3. Paleontologists deal with

   a. Embryological evidences
   b. Fossil evidences
   c. Vestigial organ evidences
   d. All the above

4. The best way of direct dating fossils of recent origin is by

   a. Radio-carbon method
   b. Uranium lead method
   c. Potassium-argon method
   d. Both (a) and (c)

5. The term Ethnobotany was coined by

   a. Khorana  b. J.W. Harsbberger
   c. Ronald Ross  d. Hugo de Vries

## II Fill in the blanks

1. The characters developed by the animals during their life time, in response to the environmental changes are called _____.
2. The degenerated and non-functional organs found in an organism are called _____.
3. The forelimbs of bat and human are examples of _____ organs.
4. The theory of natural selection for evolution was proposed by _____.

## III State true or false. Correct the false statements

1. The use and disuse theory of organs' was postulated by Charles Darwin.
2. The homologous organs look similar and perform similar functions but they have different origin and developmental pattern.
3. Birds have evolved from reptiles.

## IV Match the following

| | Column A | Column B |
|---|---|---|
| a) | Atavism | caudal vertebrae and vermiform appendix |
| b) | Vestigial organs | a forelimb of a cat and a bat's wing |
| c) | Analogous organs | rudimentary tail and thick hair on the body |
| d) | Homologous organs | a wing of a bat and a wing of an insect |
| e) | Wood park | radiocarbon dating |
| f) | W.F. Libby | Thiruvakkarai |

## V Answer in a word or sentence

1. A human hand, a front leg of a cat, a front flipper of a whale and a bat's wing look dissimilar and adapted for different functions. What is the name given to these organs?
2. Which organism is considered to be the fossil bird?
3. What is the study of fossils called?

## VI Short answers questions

1. The degenerated wing of a kiwi is an acquired character. Why is it an acquired character?
2. Why is Archaeopteryx considered to be a connecting link?
3. Define Ethnobotany and write its importance.
4. How can you determine the age of the fossils?

## VII Long answer questions

1. Natural selection is a driving force for evolution-How?
2. How do you differentiate homologous organs from analogous organs?
3. How does fossilization occur in plants?



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